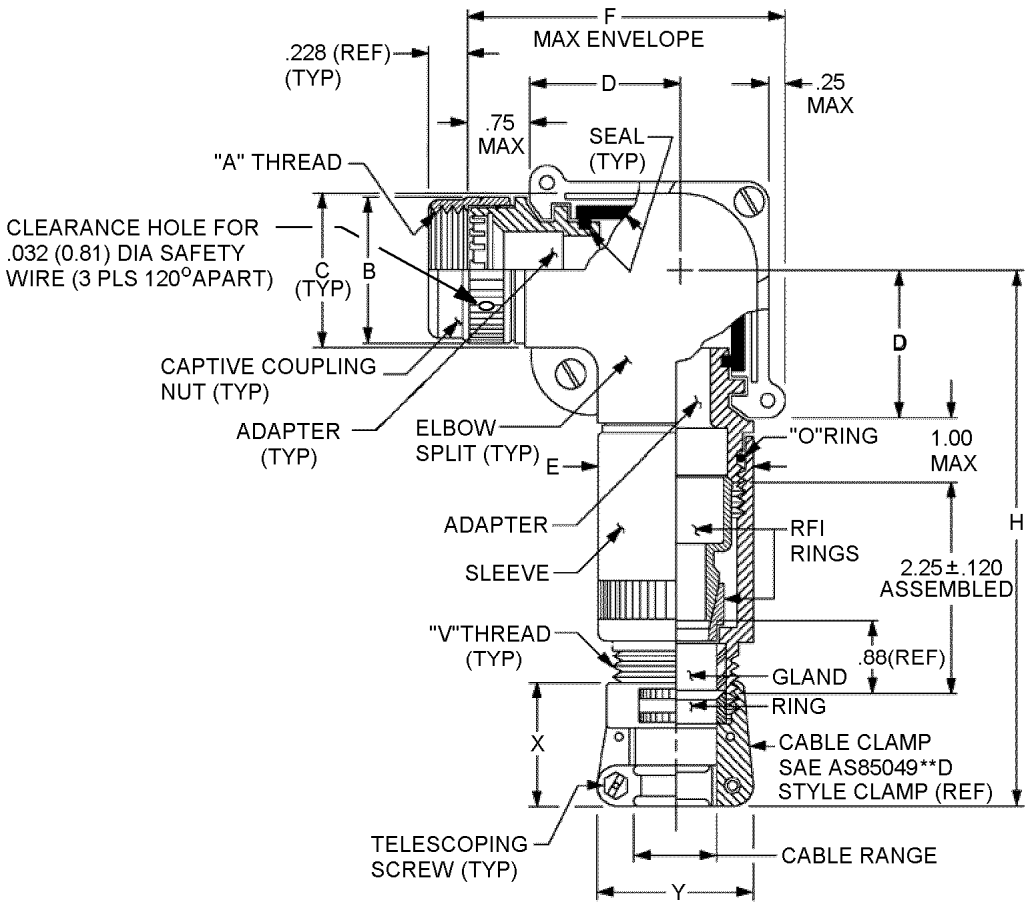
INCH-POUND
MIL-DTL-28840/8E
13 May 2009
SUPERSEDING
MIL-DTL-28840/8D
26 April 2007

DETAIL SPECIFICATION SHEET

CONNECTORS, ELECTRICAL, CIRCULAR, THREADED, HIGH SHOCK, HIGH DENSITY, SHIPBOARD, BACKSHELL, 90°, CABLE SEALING AND SHIELD TERMINATION

This specification is approved for use by all Departments and Agencies of the Department of Defense.

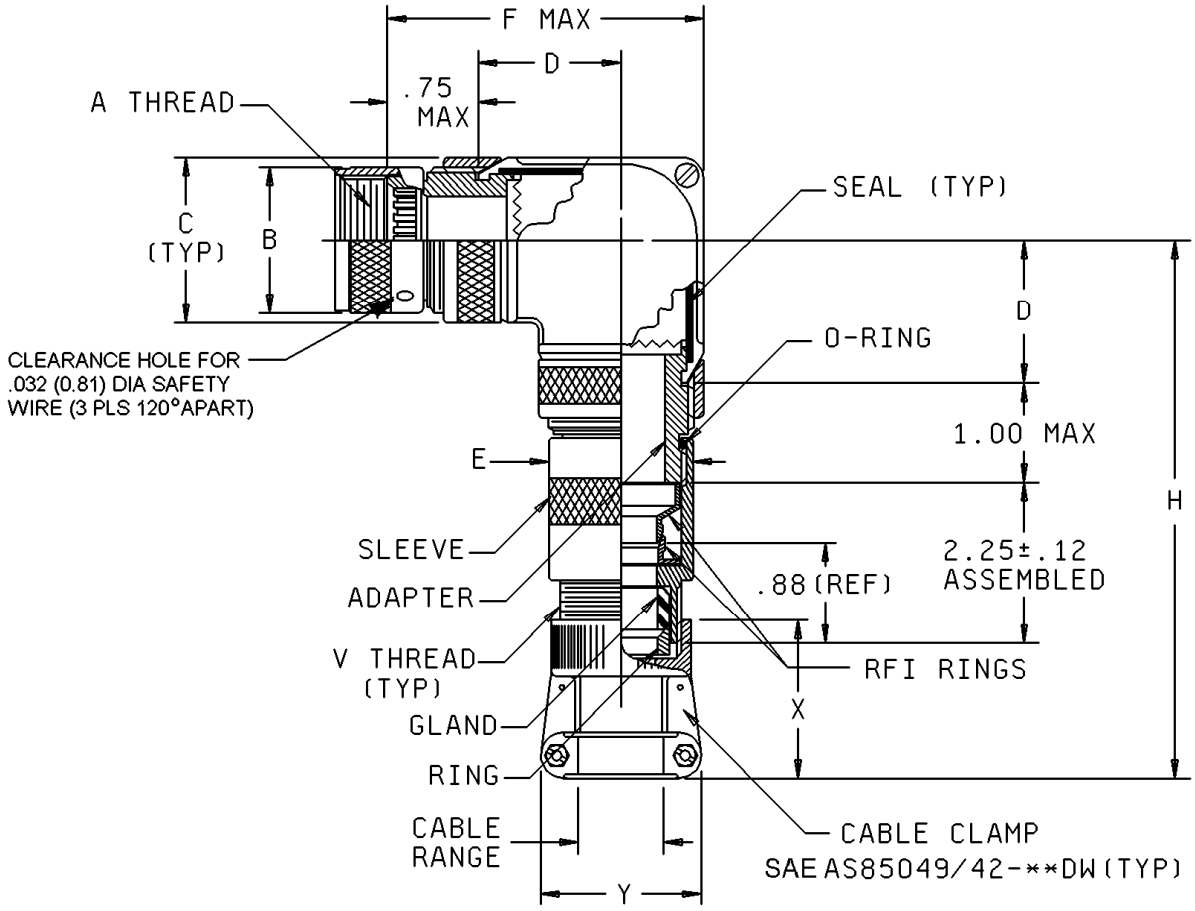
The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-28840.



STYLE A

FIGURE 1. Dimensions and configurations.

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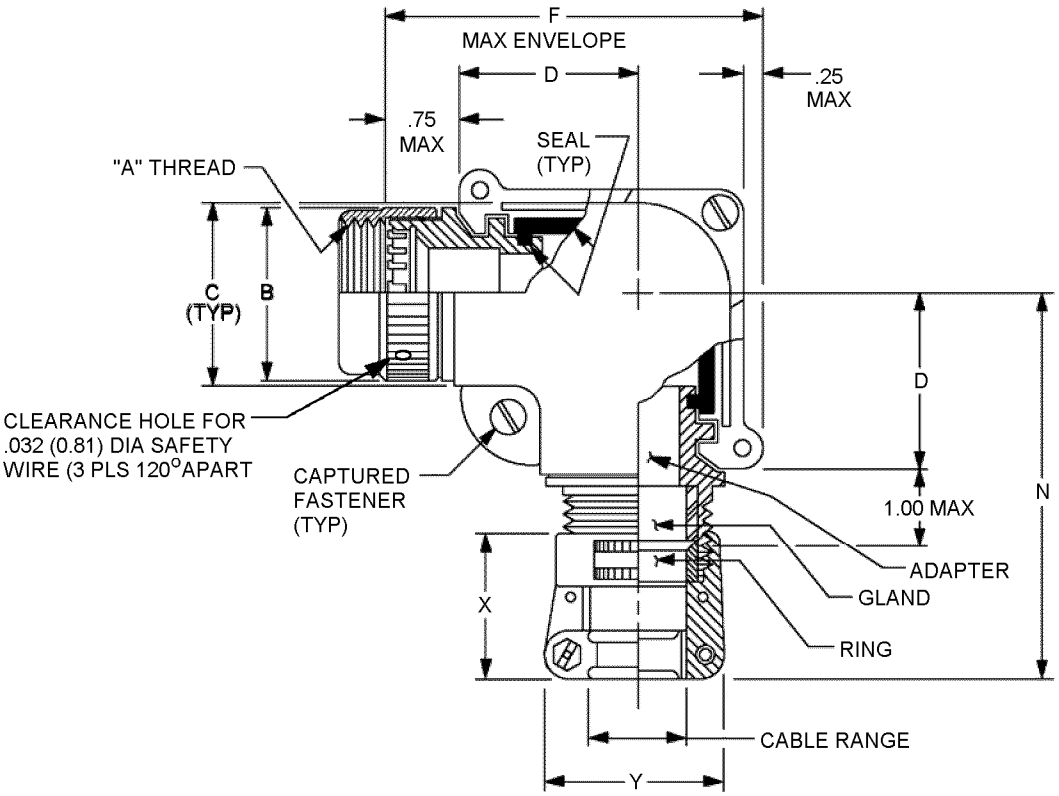


ALTERNATE STYLE A

Inches	mm
.12	3.05
.228	5.79
.25	6.35
.75	19.05
.88	22.35
2.25	57.15

FIGURE 1. Dimensions and configurations - Continued.

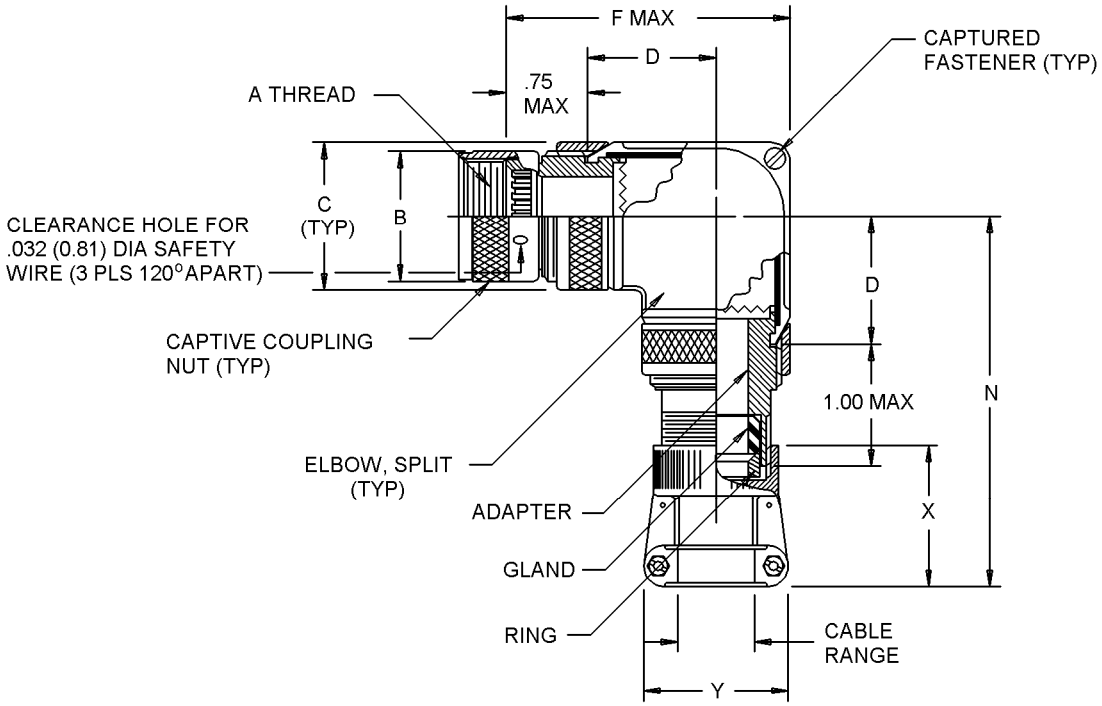
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STYLE B

FIGURE 1. Dimensions and configurations - Continued.

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ALTERNATE STYLE B

FIGURE 1. Dimensions and configurations - Continued.

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Backshell size (see note 3)	Dash no.	Designator (see note 5)	A thread class 2B	B max dia	C max dia	D max dim
A	01	A (11)	.750-20 UNEF	1.028 (26.11)	1.188 (30.17)	1.031 (26.19
A	02	B (13)	.875-20 UNEF	1.141 (28.98)		1.031 (26.19)
B	03	B (13)				
A	04	C (15)	1.00-20 UNEF	1.263 (32.08)	1.438 (36.52)	1.156 (29.36)
B	05	C (15)				
C	06	C (15)				
D	07	C (15)			1.688 (42.87)	1.281 (32.54)
A	08	D (17)	1.125-18 UNEF	1.387 (35.23)		
B	09	D (17)				
C	10	D (17)				
A	11	E (19)	1.250-18UNEF	1.513 (38.43)		
B	12	E (19)			1.938 (49.22)	1.406 (35.71)
A	13	F (23)	1.438-18 UNEF	1.703 (43.51)		
B	14	F (23)				
C	15	F (23)				
A	16	G (25)	1.562-18 UNEF	1.825 (46.35)		
B	17	G (25)				
C	18	G (25)			2.188 (54.43)	1.531 (38.89)
D	19	G (25)				
A	20	H (29)	1.875-16 UN	2.143 (54.43)		
B	21	H (29)				

FIGURE 1. Dimensions and configurations - Continued.

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Back shell size (see note 3)	Dash no.	Designator (see note 5)	Cable range		E max dia	F max dim	V thread class 2* (see note 6)
			Max	Min			
A	01	A (11)	.312 (7.92)	.188 (4.77)	1.156 (29.36)	2.614 (66.40)	.625-24 UNEF
A	02	B (13)	.312 (7.92)	.188 (4.77)	1.156 (29.36)		.625-24 UNEF
B	03	B (13)	.438 (11.12)	.281 (7.14)			.750-20 UNEF
A	04	C (15)	.312 (7.92)	.188 (4.77)	1.281 (32.54)	2.864 (72.75)	.625-24 UNEF
B	05	C (15)	.438(11.12)	.281 (7.14)			.750-20 UNEF
C	06	C (15)	.562 (14.27)	.344 (8.74)			.875-20 UNEF
D	07	C (15)	.750 (19.05)	.438 (11.12)	1.406 (35.71)	3.114 (79.10)	1.188-18 UNEF
A	08	D (17)	.625 (15.87)	.375 (9.82)			1.000-20 UNEF
B	09	D (17)	.750 (19.05)	.438 (11.12)			1.188-18 UNEF
C	10	D (17)	.938 (23.82)	.625 (15.87)	1.531 (38.89)		1.438-18 UNEF
A	11	E (19)	.625 (15.87)	.375 (9.52)	1.406 (35.71)		1.000-20 UNEF
B	12	E (19)	.938 (23.82)	.625 (15.87)	1.531 (38.89)	3.364 (85.45)	1.438-18 UNEF
A	13	F (23)	.750 (19.05)	.438 (11.12)	1.406 (25.71)		1.188-18 UNEF
B	14	F (23)	.938 (23.82)	.625 (15.87)	1.531 (38.89)		1.438-18 UNEF
C	15	F (23)	1.188 (30.17)	.812(20.62)	1.781 (45.24)		1.750-18 UNS
A	16	G (25)	.938 (23.82)	.625 (15.87)	1.531 (38.89)		1.438-18 UNEF
B	17	G (25)	1.188 (30.17)	.812 (20.62)	1.781 (45.24)		1.750-18 UNS
C	18	G (25)	1.250 (31.75)	.875 (22.23)		3.614 (91.80)	1.750-18 UNS
D	19	G (25)	1.380 (35.05)	1.000 (25.40)	2.156 (54.76)		2.000-16 UNS
A	20	H (29)	1.188 (30.17)	.812 (20.62)	1.781 (45.24)		1.750-18 UNS
B	21	H (29)	1.380 (35.05)	1.000 (25.40)	2.156 (54.76)		2.000-18 UNS

FIGURE 1. Dimensions and configurations - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T19:34Z
Check the source to verify that this is the current version before use.

Back shell size (see note 3)	Dash no.	Designator (see note 5)	X ref dim	Y ref dim	H max style A	N max style B
A	01	A (11)	1.281 (32.54)	.938 (23.82)	5.332 (134.43)	2.962 (75.23)
A	02	B (13)	1.281 (32.54)	.938 (23.82)	5.332 (134.43)	2.962 (75.23)
B	03	B (13)	1.281 (32.54)	1.125 (28.57)	5.332 (134.43)	2.962 (75.23)
A	04	C (15)	1.281 (32.54)	.938 (23.82)	5.457 (138.61)	3.087 (78.41)
B	05	C (15)	1.281 (32.54)	1.125 (28.57)	5.457 (138.61)	3.087 (78.41)
C	06	C (15)	1.281 (32.54)	1.312 (33.32)	5.457 (138.61)	3.087 (78.41)
D	07	C (15)	1.281 (32.54)	1.593 (40.62)	5.682 (144.32)	3.312 (84.12)
A	08	D (17)	1.281 (32.54)	1.312 (33.32)	5.682 (144.32)	3.312 (84.12)
B	09	D (17)	1.312 (33.32)	1.593 (40.62)	5.713 (145.11)	3.343 (84.91)
C	10	D (17)	1.406 (35.71)	1.750 (44.45)	5.807 (147.50)	3.437 (87.30)
A	11	E (19)	1.281 (32.54)	1.312 (33.32)	5.682 (144.32)	3.312 (84.12)
B	12	E (19)	1.406 (35.71)	1.750 (44.45)	5.832 (148.13)	3.462 (87.93)
A	13	F (23)	1.312 (33.32)	1.593 (40.62)	5.738 (145.25)	3.368 (85.55)
B	14	F (23)	1.406 (35.71)	1.750 (44.45)	5.832 (148.13)	3.462 (87.93)
C	15	F (23)	1.593 (40.46)	2.093 (53.16)	6.019 (152.88)	3.649 (92.68)
A	16	G (25)	1.406 (35.71)	1.750 (44.45)	5.832 (148.13)	3.462 (87.93)
B	17	G (25)	1.593 (40.46)	2.093 (53.16)	6.019 (152.88)	3.649 (92.68)
C	18	G (25)	1.593 (40.46)	2.093 (53.16)	6.144 (156.06)	3.774 (95.86)
D	19	G (25)	1.625 (41.28)	2.343 (59.51)	6.176 (156.87)	3.806 (96.62)
A	20	H (29)	1.593 (40.46)	2.093 (53.16)	6.144 (156.06)	3.774 (95.86)
B	21	H (29)	1.625 (41.28)	2.343 (59.51)	6.176 (156.87)	3.806 (96.62)

FIGURE 1. Dimensions and configurations - Continued.

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Back shell size (see note 3)	Dash no.	Designator (see note 5)	A thread class 2B	B max dia	C max dia	D max dia
A	22	J (33)	2.062-16 UNS	2.329 (59.16)	2.688 (68.26)	1.781 (45.24)
A	23	J (33)				
B	24	A (11)	.750-20 UNEF	1.028 (26.11)	1.188 (30.17)	1.031 (26.19)
C	25	A (11)			1.438 (36.52)	1.156 (29.36)
C	26	B (13)	.875-20 UNEF	1.141 (28.98)		
D	27	B (13)			1.688 (42.87)	1.281 (32.54)
E	28	C (15)	1.000-20 UNEF	1.263 (32.08)		

Back shell size (see note 3)	Dash no.	Designator (see note 5)	Cable range		E max dim	F max dim	V thread class 2* (see note 6)
			max	min			
A	22	J (32)	1.380 (35.05)	1.000 (25.40)	2.281 (57.94)	4.114 (104.50)	2.000-18 UNS
A	23	J (33)	1.625 (41.26)	1.250 (31.75)			2.250-16 UNS
B	24	A (11)	.438 (11.12)	.281 (7.14)	1.156 (29.36)	2.614 (66.40)	.750-20 UNEF
C	25	A (11)	.562 (14.27)	.344 (8.74)	1.281 (32.54)	2.864 (72.75)	.875-20 UEF
C	26	B (13)	.562 (14.27)	.344 (8.74)	1.281 (32.54)		.875-20 UEF
D	27	B (13)	.625 (15.87)	.375 (9.52)	1.406 (35.71)	3.114 (79.10)	1.000-20 UNEF
E	28	C (15)	.625 (15.87)	.375 (9.50)			1.000-20 UNEF

FIGURE 1. Dimensions and configurations - Continued.

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Back shell size (see note 3)	Dash no.	Designator (see note 5)	X ref dim	Y ref dim	H max style A	N max style B
A	22	J (33)	1.625 (41.28)	2.343 (59.51)	6.426 (163.22)	4.056 (103.02)
B	23	J (33)	1.900 (48.26)	2.750 (69.85)	6.701 (170.21)	4.331 (110.01)
B	24	A (11)	1.281 (32.54)	1.125 (28.57)	5.332 (135.43)	2.962 (75.23)
C	25	A (11)	1.281 (32.54)	1.312 (33.32)	5.457 (138.61)	3.087 (78.41)
C	26	B (13)	1.281 (32.54)	1.312 (33.32)	5.457 (138.61)	3.087 (78.41)
D	27	B (13)	1.281 (32.54)	1.312 (33.32)	5.682 (144.32)	3.312 (84.12)
E	28	C (15)	1.281 (32.54)	1.312 (33.32)	5.682 (144.32)	3.312 (84.12)

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Designator of backshell size within each shell size.
4. Dimensions apply after plating.
5. Shell sizes are provided within parentheses for information and are not a part of the designator.
6. See MIL-DTL-28840 for V thread, class 2*.

FIGURE 1. Dimensions and configurations - Continued.

REQUIREMENTS

Dimensions and configurations: See figure 1.
Interface dimensions shall conform to MIL-DTL-28840.

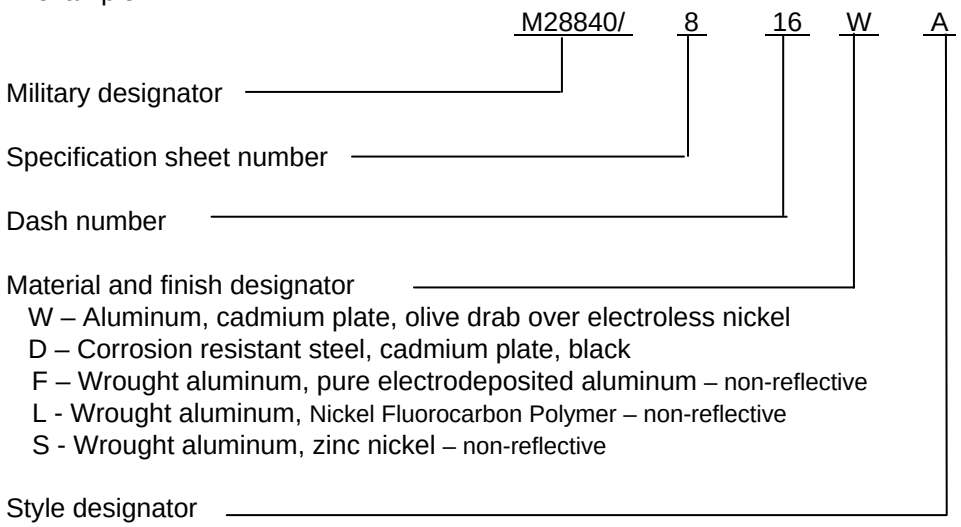
Material: Shall be aluminum alloy in accordance with ASTM B85, ASTM B108/B108M, ASTM B26/B26M, ASTM B209, ASTM B211 or ASTM B221. Corrosion resistant steel shall be in accordance with ASTM A276, ASTM A580/A580M or ASTM A473, type 316.

Finish: Aluminum components shall plated in accordance with MIL-DTL-28840. The screws and washers shall be passivated in accordance with SAE-AMS2700, type 2.

Marking: The backshell shall be identified with the manufacturer's name or trademark and the Part or Identifying Number (PIN).

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PIN example:



Qualification: Two samples in each shell size for which qualification is desired and shall be tested in accordance with MIL-DTL-28840.

Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Referenced documents. In addition to MIL-DTL-28840, this document references the following:

ASTM A276
ASTM A473
ASTM A580/A580M
ASTM B26/B26M
ASTM B85
ASTM B108/B108M
ASTM B209
ASTM B211
ASTM B221
SAE-AMS2700

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CONCLUDING MATERIAL

Custodians:
Navy - EC
DLA - CC

Preparing activity:
DLA - CC

(Project 5935-2007-220)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

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